Cover Letter — Submission to Remote Sensing of Environment

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[Date of submission]

The Editors-in-Chief *Remote Sensing of Environment* Elsevier B.V.

Dear Editors,

I am pleased to submit our original research article entitled **“Decoupling Cyclone-Induced Saline Inundation from Agronomic Flooding in Sentinel-1/2 Rice Phenology Retrieval: A Multi-Year Bay-of-Bengal Coastal Framework (2017–2024)”** for consideration in *Remote Sensing of Environment*.

Why this manuscript matters now. Almost every published Sentinel-1 rice phenology algorithm uses the SAR backscatter trough at flooding/transplanting as the anchor for start-of-season detection. Yet in cyclone-prone Asian deltas — which produce roughly 11 % of global rice — that same trough can be generated by storm-surge saline inundation weeks earlier in the season, silently corrupting derived phenological dates in precisely the years that vulnerability assessments need most. To our knowledge, no prior study has characterised, let alone corrected, this physical confound.

What we did. Using Cyclones Fani (May 2019), Bulbul (November 2019), Amphan (May 2020) and Yaas (May 2021) as natural experiments in a Before-After-Control-Impact design across five coastal Odisha districts (Balasore, Bhadrak, Kendrapara, Jagatsinghapur, Puri) with three inland controls (Dhenkanal, Angul, Cuttack), we (i) developed an open-source random-forest classifier that separates cyclone-induced saline inundation from agronomic transplanting flooding using eight Sentinel-1, Sentinel-2, JRC water, and ERA5 features; (ii) propagated the classifier output into a parallel raw vs. corrected phenology-extraction pipeline based on Whittaker-smoothed, double-logistic curve fitting; and (iii) quantified the resulting bias in SOS/POS/EOS dates with a mixed-effects BACI model. We validated using a fully open-data, multi-source strategy: against the NASA MODIS MCD12Q2 Land Surface Phenology product as the methodologically independent primary reference, against ICRISAT Village Dynamics in South Asia (VDSA) microdata for the Bhadrak benchmark site, against district-level Kharif rice yield records from data.gov.in, against PlanetScope NICFI 3-m visual interpretation at sixty stratified sites, and against two existing rice mask products (Mondal et al. 2022; Singha et al. 2019). We additionally tested transferability on the Andhra Pradesh coast under Cyclone Hudhud (October 2014). Every dataset used in the study is publicly downloadable without permission, application, or institutional gatekeeping — a deliberate design choice that maximises reviewer reproducibility and removes any data-access dependency from the publication record.

Why RSE specifically. This work fits the journal’s scope on biophysical applications of remote sensing to agriculture in three ways: (1) it removes a previously uncharacterised barrier to robust phenological retrieval rather than incrementally improving accuracy; (2) it delivers a named, openly-licensed deployable artefact (the RiceBaCI-GEE toolkit at https://github.com/pandasupranab/RiceBaCI-GEE and a 10-m corrected SOS/POS/EOS dataset on Mendeley Data); and (3) it speaks directly to the climate-vulnerability and parametric-insurance communities that the *RSE* readership serves. The methodology was pre-registered on the Open Science Framework prior to data analysis (https://osf.io/[OSF-id-pending]), and all code is publicly available under MIT licence.

Originality and ethics. The manuscript has not been published elsewhere and is not under consideration at another journal. All authors approve this submission and have made substantial intellectual contributions consistent with the journal's authorship policy. We declare no competing interests, no funding to disclose, and we report the use of generative-AI assistants in the manuscript as required by Elsevier policy.

Suggested reviewers. We propose the following experts whose recent work most directly bears on our contribution; none is a current or recent collaborator:

1. **Dr. Mirco Boschetti** — CNR-IREA, Milan, Italy (mirco.boschetti@irea.cnr.it). Lead author of the PhenoRice algorithm — the most cited rice phenology retrieval framework; his expertise is directly relevant to our SOS/POS/EOS double-logistic methodology.

2. **Dr. Mrinal Singha** — Indian Institute for Human Settlements, Bengaluru, India (mrinal.singha@iihs.ac.in). Author of the South-Asia Sentinel-1 paddy rice product (Singha et al. 2019, *Scientific Data*) and the Bangladesh flood-affected rice paper (Singha et al. 2020, *ISPRS J P&RS*) — both cited in our manuscript.

3. **Dr. Nguyen Lam-Dao** — Vietnam National Space Center, VAST, Ho Chi Minh City, Vietnam (nldao@vnsc.org.vn). Lead author of the Mekong-Delta Sentinel-1 salinity-rice mapping (Hoang-Phi et al. 2020, *Remote Sensing*) — directly comparable saline-inundation-rice problem in a different delta, providing methodological cross-check.

4. **Dr. Saptarshi Mondal** — VassarLabs / Birla Institute of Technology Mesra, India (saptarshi.mondal@vassarlabs.com). Author of the Odisha rice-fallow Sentinel-1 work (Chandna and Mondal 2020, *GIScience & Remote Sensing*) — eastern Indian rice remote-sensing expertise on the same study area.

5. **Prof. Clement Atzberger** — University of Natural Resources and Life Sciences (BOKU), Vienna, Austria (clement.atzberger@boku.ac.at). Senior authority on agricultural remote sensing time series; *RSE* contributor with broad methodological expertise.

We have no reviewers to exclude due to conflict of interest.

I thank the Editors and reviewers in advance for their consideration of our manuscript.

Yours sincerely,

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3. Have your supervisor read the cover letter; their initials at the bottom strengthen credibility.
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